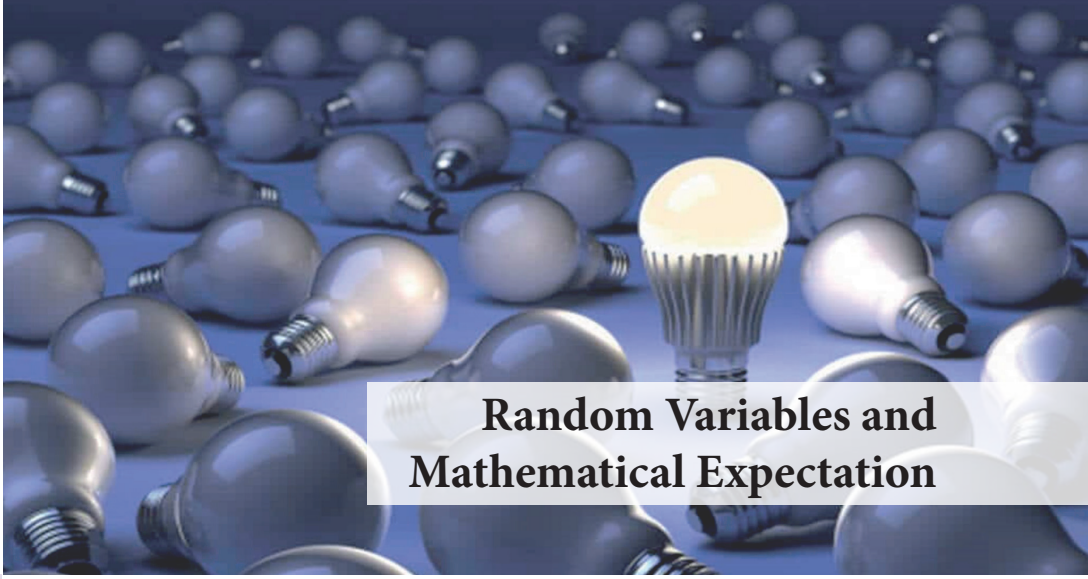
Chapter

9



Random Variables and Mathematical Expectation



Siméon-Denis Poisson
(21 June, 1781 – 25 April, 1840)

Siméon-Denis Poisson, a French mathematician known for his work on definite integrals, electromagnetic theory, and probability.

Poisson's research was mainly on Probability. Poisson distribution was one of the important invention of Poisson. Poisson contributed to the law of large numbers and it was useful for the approximation of binomial distribution. Poisson distribution is now fundamental in the analysis of problems concerning radioactivity, traffic, and the random occurrence of events in time or space.

'Everything existing in the universe is the fruit of Chance'.

- Democritus

Learning Objectives

- Knows the relevance of Random Variables.
- Understands the types of Random Variables.
- Differentiates the various Types of probability functions.
- Enumerates the functions and properties of Cumulative Distribution.
- Understands the Concept of expectation.
- Applies the Theorems on Expectations.
- Identifies Moment generating functions and Characteristic Functions

BCPX3Z